

GENERAL THETA FOUNDATIONS I: CAUSAL WIDTH BEYOND CUTWISE POSITIVE RANK

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ABSTRACT. We compare the least memory at individual cuts with the least peak memory of one exact finite-horizon stochastic transducer. We construct a uniformly full-support family on fixed alphabets whose separate minimum is three at every internal cut, whereas its minimum simultaneous width tends to infinity. The lower bound allows arbitrary time-dependent stochastic rows and hidden continuations outside the observable affine space. Its mechanism is a volume budget for projections of reachable simplex sections: each command forces an enlargement, while bounded state number bounds the number of projected vertices. For a golden-angle rotation we obtain the explicit bound $W_N \geq \frac{1}{6} \log_2(N/24000000)$, alongside an exact $4\lceil\sqrt{N} + 1\rceil$ -state construction. A rational rotation gives the same unbounded incompatibility with effective algebraic horizon thresholds and rational finite realizations. A separate composition theorem identifies feasible profiles under a verifiable retained tag as Minkowski sums; applied to the two-cut obstruction, it gives the exact frontier $K_1, K_2 \geq 3m$, $K_1 + K_2 \geq 7m$. The finite-horizon volume inequality, rather than a stationary nonrealizability statement, is the central relation between local positive rank and causal memory.

1. INTRODUCTION

A stochastic response array can have small positive factorizations at every cut without admitting one machine that realizes those factorizations in succession. The issue is not only how many continuations enclose the observed rows. Every chosen continuation must also evolve by the same legal stochastic update as the other continuations at that time. We study the resulting difference between *separate* and *simultaneous* state complexity.

Our principal result gives an unbounded difference with fixed input and output alphabets. Fix $a = 1/10$, a planar rotation R , a seed $x \in \{-1, 1\}^2$, a word $c = (c_1, \dots, c_N) \in \{0, 1\}^N$, and a query $j \in \{1, 2\}$. The seed is read first, the commands are read in order, and the query is supplied last. The required binary response is

$$(1) \quad C_{N,R}(b \mid x, c, j) = \frac{1 + b e_j^T R^{c_1 + \dots + c_N} a x}{2}, \quad b \in \{-1, 1\}.$$

The same formula is required for every input word, not merely under a chosen distribution on words. Every entry lies between $(1 - \sqrt{2}/10)/2$ and $(1 + \sqrt{2}/10)/2$.

At any one internal cut, three normalized continuations suffice and are necessary. This remains true when the cut is required to belong to a complete machine and all its other registers may be enlarged. Write $W_N(R)$ for the least peak over machines satisfying all cuts at once.

Theorem 1.1 (Fixed local rank and growing causal width). *Let $\alpha = (\sqrt{5} - 1)/2$ and let R rotate by $2\pi\alpha$. Every internal cut of (1) has separate minimum three and ordinary normalized*

Date: September 25, 2026. Revision 32.

2020 Mathematics Subject Classification. Primary 93B15, 68Q45; Secondary 52A20, 60J10, 62B05.

Key words and phrases. Positive realization, finite stochastic transducers, causal memory, polyhedral lifts, rotation, compatible factorizations.

residual rank three. For $N \geq 1$,

$$(2) \quad \max \left\{ 3, \frac{1}{6} \log_2 \frac{N}{24000000} \right\} \leq W_N(R) \leq \min \left\{ 3(N+1), 4 \lceil \sqrt{N+1} \rceil \right\}.$$

All lower bounds allow arbitrary randomized encoders, arbitrary hidden state continuations, and cut-dependent stochastic transition matrices. The upper bounds realize every conditional probability exactly.

For the rational rotation

$$(3) \quad R_0 = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix},$$

the same separate minima and upper bounds hold and $W_N(R_0) \rightarrow \infty$. For each proposed state bound, an excluded horizon can be computed by real algebraic optimization. The $3(N+1)$ -state realization is rational.

The proof occupies Sections 3–5. The logarithmic lower bound and square-root upper bound are not asserted to be sharp. They show, with explicit constants in one family, that uniformly bounded separate positive ranks do not control simultaneous width even in the permissive clocked model. The rational version makes the qualitative phenomenon independent of transcendental input data.

The principal technical point is the treatment of nonminimal hidden representations. If a machine has K_t states, its hidden state vectors need not individually have an observable two-dimensional continuation. Projecting those individual states into a presumed invariant polygon would therefore restrict the competing class. Instead, let H_t be the affine span of the actually reachable state *distributions*, and intersect it with the probability simplex. Projecting $H_t \cap \Delta_{K_t}$ gives a valid observable polytope with at most 2^{K_t} vertices. Stochastic updates carry these sections forward. A command and an idle command then force a nested sequence of observable polytopes. A strictly positive volume increment, summed over time, provides the lower bound. This is a necessary argument for all hidden realizations, not a restriction to residual-state or affine-minimal machines.

A second result concerns exact finite profile composition. Theorem 6.1 proves that a verifiable retained tag makes feasible profiles add as sets. In particular, the earlier two-cut obstruction with frontier $(3, 4)$, $(4, 3)$ yields

$$K_1 \geq 3m, \quad K_2 \geq 3m, \quad K_1 + K_2 \geq 7m.$$

Its optimum peak is $\lceil 7m/2 \rceil$, whereas either cut separately has minimum $3m$. This closure law complements the temporal example: it identifies an exact growing Pareto frontier, while the rotation family keeps alphabets and all separate minima fixed. The tagged construction has structural output zeros; it is not called a full-support example.

1.1. Classical mechanisms and the new comparison. Invariant polyhedral cones and normalized convex sections are classical in positive realization [6, 1, 12]. Irrational rotation as an obstruction to a finite *stationary* positive realization is also classical: see [1, Example 4] and [8, Theorem 6, Example 2 and Lemma 7]. We do not claim either mechanism as a new discovery. Reachable subspaces and observable quotients are likewise established tools [8, Section 4].

The comparison proved here differs in its quantifiers. Every finite horizon has a finite exact realization; both the state spaces and the rows may change at every epoch. There is no common stationary matrix to which a peripheral-spectrum obstruction can be applied. Moreover, the least *positive* factorization at each cut, not just ordinary rank, is three. The cumulative volume bound and the explicit sequence of finite realizations concern this finite, time-dependent problem. Section 7 details the relation to positive realization, nonnegative factorization and residual

automata. The explicit bounds are claims of the present article; exhaustive priority clearance is not inferred from the bibliography.

Statement	Mathematical role
Reachable-section and volume budget	General multi-cut lower bound; Sections 3–5
Rotation family	Fixed-rank, fixed-alphabet unbounded incompatibility
Tagged profile sum	Exact composition of feasible profile sets
Two-cut primitive	Inherited from revision 31; proof retained in Appendix A
Invariant cones and weighted replacement	Classical mechanisms; not new claims

2. COMPLETE ARRAYS AND CHARGED STATE PROFILES

Definition 2.1 (Clocked stochastic transducer). Fix finite alphabets $X, Y_1, \dots, Y_N, J, B$. A complete terminal array is a probability distribution $C(\cdot \mid x, y_1, \dots, y_N, j)$ on B for every word in the full Cartesian product. A realization has finite registers $S_t = [K_t]$ for $0 \leq t \leq N$, an initialization row $E(s \mid x)$, transition rows $T_{t,y}(s, s')$ from S_t to S_{t+1} for $0 \leq t < N$, and terminal rows $D(b \mid s, j)$, all nonnegative and normalized. It realizes C when

$$(4) \quad C(b \mid x, y_1, \dots, y_N, j) = \sum_{s_0, \dots, s_N} E(s_0 \mid x) \prod_{t=0}^{N-1} T_{t, y_{t+1}}(s_t, s_{t+1}) D(b \mid s_N, j).$$

The current input is read atomically, and only the next register survives its update. There is an initial singleton before x and, when the output is retained, a final $|B|$ -label register. The full peak is $\max(1, |B|, K_0, \dots, K_N)$.

Inputs in this definition are externally supplied; they need not have a probability law. Equality is wordwise. A controller’s earlier randomness, input buffers, data-dependent scheduling information and previously emitted symbols cannot be read again unless retained in its register. The declared epoch t is a free external clock; the current query is consumed in the terminal update. All row tables are known program data, and sampling a stochastic row is atomic. Table size, arithmetic work, construction time and fair-bit sampling workspace are not included in K_t . A shared private seed is not an additional free input.

We count available labels. Unreachable labels may be removed, while padding by unused labels explains upward closure of feasible profiles. Every declared input row, including rows unused by the actual initial law, is a stochastic row. An autonomous implementation instead uses one state set and time-independent rows; phase tagging gives a sufficient sum-of-widths bound and is not identified with the clocked peak. These conventions separate the finite positive-realization problem from succinct streaming-program or bit-operation complexity.

Let $\mathcal{F}(C) \subset \mathbb{N}^{N+1}$ be the upward-closed set of realizable profiles. For a prefix $h = (x, y_1, \dots, y_t)$, let R_h be its complete conditional array on all suffixes and terminal queries. At this cut the ambient continuation polytope is a product of output simplices. Define $r_t = \text{rank}(R_h)_h$, with columns indexed by suffix, query and output. Normalized rows give $r_t = 1 + \dim \text{aff}\{R_h\}$.

Definition 2.2 (Separate minimum). The separate minimum $\kappa_t(C)$ is the least k such that every R_h is a convex combination of k normalized continuation arrays on the same future alphabet. There is no compatibility requirement between the dictionaries selected at distinct cuts. The minimum peak of a compatible realization is denoted $W(C)$.

For finite alphabets, $\kappa_t(C) = \min_{\mathbf{K} \in \mathcal{F}(C)} K_t$: store the whole prefix before the chosen cut, implement its encoder at that cut, and afterwards store the selected label and future inputs until the terminal update. Thus a separate factorization can always be embedded in a complete

machine, at possibly large costs elsewhere. Ordinary row rank is a lower bound on κ_t . For a single terminal output alphabet this minimum is ordinary nonnegative rank after normalization. For a queried continuation array, every suffix/query block of a generator must normalize separately; nonnegative rank of the concatenated matrix need only be a lower bound. These are not the same as residual-only state counts [5, 3].

Notation	Meaning
$K_t, W(C)$	available labels at a cut; least simultaneous peak
$r_t, \kappa_t(C)$	ordinary row rank; separate normalized positive minimum
E_h, H_t	reachable state distribution; its affine span
Q_t, P_t	reachable simplex section; observable projected section
S, D	compulsory seed hull; bounded observable domain
δ_M	minimum one-step volume expansion for at most M vertices

The input-only model is already a subclass of finite causal experiments: a prescribed action alphabet may consist of one action at each stage. A lower bound in this class does not solve optimization over informative experimental actions. The present advance concerns arbitrary-horizon compatibility of exact responses under a specified interface.

3. REACHABLE SECTIONS AND A CUMULATIVE VOLUME BOUND

We first treat a general finite collection of linear mean updates. Let $\mathcal{L}$ be a finite set of linear maps of $\mathbb{R}^d$, containing I_d . A finite seed alphabet has associated vectors $z(x)$, whose convex hull S has nonempty interior. Assume that every product of at most N allowed maps sends the seed vectors into $D = [-1, 1]^d$. After reading x and N commands $L_1, \dots, L_N \in \mathcal{L}$, a terminal query $j \in [d]$ requires a binary output of mean $e_j^T L_N \cdots L_1 z(x)$. Denote this complete array by C_N .

Lemma 3.1 (The reachable-section lift). *Every realization of C_N with profile $(K_t)_{t=0}^N$ determines full-dimensional polytopes $P_t \subset D$ such that*

$$(5) \quad S \subset P_t, \quad |\text{ext } P_t| \leq 2^{K_t}, \quad LP_t \subset P_{t+1} \quad (L \in \mathcal{L}, t < N).$$

The conclusion permits arbitrary hidden states and does not require individual state continuations to belong to the residual affine span.

Proof. For each actual prefix h at cut t , let $E_h \in \Delta_{K_t}$ be its hidden state distribution. Put

$$H_t = \text{aff}\{E_h : h \text{ a prefix at } t\}, \quad Q_t = H_t \cap \Delta_{K_t}.$$

For any state distribution $e \in \Delta_{K_t}$, run the remaining commands all equal to I_d . The vector of its d terminal means is a linear function $\pi_t(e)$ and belongs to D . Correctness on actual prefixes gives $\pi_t(E_h) = z_h$, the specified predictive vector. Set $P_t = \pi_t(Q_t)$.

Write $T_{t,L}$ for the realizing transition matrix of command L . The identity $E_h T_{t,L} = E_{hL}$ implies $H_t T_{t,L} \subset H_{t+1}$. Stochasticity also implies $\Delta_{K_t} T_{t,L} \subset \Delta_{K_{t+1}}$; consequently $Q_t T_{t,L} \subset Q_{t+1}$. On every actual prefix,

$$\pi_{t+1}(E_h T_{t,L}) = L \pi_t(E_h).$$

Both sides are affine in E_h , so this equality extends to all of H_t , including its points with some negative coordinates. Restricting back to Q_t yields $LP_t \subset P_{t+1}$. Taking the idle command shows $P_t \subset P_{t+1}$. The initial section contains all seeds, hence so does every P_t .

It remains to bound vertices without assuming a minimal hidden representation. In its affine hull, Q_t is a bounded polytope defined by the K_t coordinate inequalities $e_s \geq 0$. Two distinct vertices cannot have the same zero-coordinate set. Indeed, if they did, their difference would give a nonzero direction preserving all active equalities, and sufficiently small motion in both directions from either point would preserve the remaining strict inequalities. That point would

not be a vertex. There are at most 2^{K_t} zero sets. A linear image of a polytope is the convex hull of the images of its vertices, proving $|\text{ext } P_t| \leq |\text{ext } Q_t| \leq 2^{K_t}$. $\square$

The section Q_t , rather than the entire hidden simplex, is essential. For an arbitrary unit hidden state, the identity relating the two probe maps need not hold. It holds on the affine span of reachable *distributions*. This is the finite-horizon version of the accessible-space/observable-quotient precaution in classical realization [8, Section 4]; the vertex bound accounts for the resulting lift. In particular, replacing 2^{K_t} by K_t without further hypotheses would not be justified.

For a full-dimensional compact seed hull $S \subset D$, define

$$(6) \quad \delta_M(S, D, \mathcal{L}) = \min_{\substack{S \subset P \subset D \\ P \text{ a polytope, } |\text{ext } P| \leq M}} \left\{ \text{vol}_d \left(\text{conv} \bigcup_{L \in \mathcal{L}} LP \right) - \text{vol}_d(P) \right\}.$$

If the feasible set is empty, put $\delta_M = +\infty$. The maps in the union need not send all of D into D . Only the implementing sequence from Lemma 3.1 must obey that constraint. The quantity is a geometric optimization on observable polytopes, not an optimization over candidate transition tables.

Theorem 3.2 (Volume budget for a state profile). *Suppose $|\det L| = 1$ for every $L \in \mathcal{L}$ and $I_d \in \mathcal{L}$. If there is no full-dimensional polytope P with $S \subset P \subset D$ and $LP = P$ for all $L \in \mathcal{L}$, then every nonempty minimum in (6) is strictly positive. Every realization of C_N satisfies*

$$(7) \quad \sum_{t=0}^{N-1} \delta_{2^{K_t}}(S, D, \mathcal{L}) \leq \text{vol}_d(D) - \text{vol}_d(S).$$

In particular, a uniform state bound K implies $N\delta_{2^K} \leq \text{vol}_d(D) - \text{vol}_d(S)$.

Proof. The set of feasible polytopes with at most M vertices is compact in the Hausdorff metric: represent them by M points in the compact set D , permitting repetitions, and impose the closed condition that their convex hull contain S . Convex hull, finite union followed by convex hull, and volume of compact convex sets are continuous. Thus the minimum is attained whenever the set is nonempty.

Because the identity is allowed, the displayed volume difference is nonnegative. If it vanishes at P , the full-dimensional convex bodies P and $\text{conv} \bigcup_L LP$ have equal volume and one contains the other, so they are equal. Thus $LP \subset P$. The determinant hypothesis makes their volumes equal, hence $LP = P$ for every L , contrary to the assumption. Therefore $\delta_M > 0$.

For the polytopes in Lemma 3.1,

$$\text{vol}_d(P_{t+1}) - \text{vol}_d(P_t) \geq \text{vol}_d \left(\text{conv} \bigcup_L LP_t \right) - \text{vol}_d(P_t) \geq \delta_{2^{K_t}}.$$

Summing telescopes. The endpoint inclusions give (7). The minima are nonincreasing in M , which gives the assertion for a uniform bound. $\square$

The volume hypothesis is a property of the mean dynamics and the observation domain, not a small-ball or metric-dimension assumption. Under an invertible affine change of predictive coordinates, using the conjugate affine maps, both sides of (7) are multiplied by the same absolute determinant. Thus the normalized budget is independent of this coordinate choice. The proof also works for volume-preserving affine maps with an idle map.

Corollary 3.3 (Effective finite obstructions). *In dimension two, for explicitly given rational or real-algebraic polygons S, D and volume-preserving linear maps, every nonempty δ_M is a computable real-algebraic number. If the hypothesis of Theorem 3.2 holds, one can compute a*

rational $0 < c_M \leq \delta_M$ and hence a finite horizon excluding any specified uniform width K with $M = 2^K$.

Proof. Use M pairs of vertex coordinates in D as variables. Containment of each vertex of S in their hull is expressed with nonnegative barycentric variables. A finite partition by determinant signs and cyclic orders identifies each possible convex hull, including repetitions and collinear points. On every such piece the areas of P and $\text{conv} \bigcup_L LP$ are polynomial expressions, by the polygon area formula. Thus the graph of the objective, its feasible domain and the statement that a number is its minimum are first-order formulas over the real closed field of algebraic numbers. Quantifier elimination, followed by real-root isolation, computes the minimum and a positive rational lower bound when it is positive [2]. Algebraic inputs are represented by defining polynomials and isolating intervals. The procedure is finite; no polynomial complexity in K , the horizon, or a succinct circuit description is asserted. $\square$

4. A FIXED-ALPHABET FAMILY WITH UNBOUNDED COMPATIBILITY COST

We apply the volume bound with

$$a = 1/10, \quad S = a[-1, 1]^2, \quad D = [-1, 1]^2, \quad \mathcal{L} = \{I_2, R\},$$

where R is any planar rotation of infinite order. The terminal array is (1). The fixed alphabets have respectively four seed symbols, two command symbols, two queries and two outputs. Rotation preserves the norm of every actual predictive vector, so the uniform full-support bound stated in the introduction holds at every horizon.

Proposition 4.1 (All separate positive minima are three). *At every internal cut of $C_{N,R}$, ordinary normalized row rank and separate normalized positive minimum both equal three, for every rotation R and every $N \geq 0$.*

Proof. For a prefix at cut t , put $z = R^{c_1 + \dots + c_t} ax$. Its entire future response is affine in z . The all-idle prefixes contain the four seeds, which affinely span $\mathbb{R}^2$, and the all-idle suffix with the two coordinate queries distinguishes both coordinates. Normalization supplies an independent constant coordinate. Hence the residual row rank is exactly three.

Consider the triangle

$$(8) \quad T = \text{conv}\{v_0, v_1, v_2\}, \quad v_0 = (1/2, 0), \quad v_1 = (-1/2, 1/2), \quad v_2 = (-1/2, -1/2).$$

It contains the centered disk of radius $1/(2\sqrt{5})$, while every vertex has norm at most $1/\sqrt{2}$. All reachable z have norm $\sqrt{2}/10 < 1/(2\sqrt{5})$, and thus lie in T . Starting from any v_s , any future rotation word has both mean coordinates in $[-1, 1]$. It therefore defines a legal normalized future channel. Barycentric decomposition of z in T gives an exact three-state factorization of the whole suffix array. This proves $\kappa_t = 3$; Definition 2.2 embeds it in a complete machine with finite, unrestricted registers at other cuts. $\square$

Theorem 4.2 (Unbounded width with effective rational data). *For every infinite-order rotation R , $W_N(R) \rightarrow \infty$. More precisely, put $\delta_M = \delta_M(S, D, \{I, R\})$. These numbers are positive for $M \geq 3$, and every feasible profile satisfies*

$$(9) \quad \sum_{t=0}^{N-1} \delta_{2^{K_t}} \leq \frac{99}{25}.$$

For R_0 in (3), the target arrays are rational and the thresholds excluding a bounded width are effective in the sense of Corollary 3.3.

Proof. An invariant polygon containing S would have nonzero vertices whose orbits under R remain in its finite vertex set, since an invertible linear map taking a polygon onto itself permutes its vertices. An infinite-order planar rotation has no finite orbit away from zero. No such polygon exists. The triangle in (8) is feasible for the definition of δ_M when $M \geq 3$, so these minima are positive by Theorem 3.2. Its volume budget gives (9), since $4 - 4a^2 = 99/25$. For any K , horizons greater than $99/(25\delta_{2K})$ exclude all width- K machines. This proves the asserted divergence.

The eigenvalue $\zeta = (3 + 4i)/5$ of R_0 is not a root of unity. Otherwise $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer, hence an integer. The contradiction proves that R_0 has infinite order. Rationality of its powers and of the seeds proves rationality of every specified channel entry. Corollary 3.3 applies. $\square$

The divergence does not follow merely by invoking the classical absence of a stationary positive realization of an irrational rotation. The matrices of a proposed machine may change at every time, and depend on N . Inequality (9) excludes these machines as well. In particular there is no function of the maximum separate positive minimum alone that bounds simultaneous width on this class.

4.1. Constructive finite realization.

Theorem 4.3 (Exact polygon realization). *For any planar rotation R and any $N \geq 1$, the channel $C_{N,R}$ has an exact realization of peak at most $4\lceil\sqrt{N+1}\rceil$. It also has an exact realization of peak $3(N+1)$, with rational rows when R has rational entries. Both constructions store finite labels, not real predictive vectors.*

Proof. Put $M = 4\lceil\sqrt{N+1}\rceil$, $u_s = (\cos(2\pi s/M), \sin(2\pi s/M))$ for $0 \leq s < M$, and

$$\rho_t = \frac{\sqrt{2}}{10} \sec^{t+1}(\pi/M), \quad P_t = \text{conv}\{\rho_t u_s : 0 \leq s < M\} \quad (0 \leq t \leq N).$$

The inradius of P_0 is $\sqrt{2}/10$, so $S \subset P_0$. The inradius of P_{t+1} is ρ_t , and hence

$$(10) \quad P_t \cup RP_t \subset P_{t+1}.$$

All these polygons lie in D . Indeed, with $L = N + 1$,

$$\cos(\pi/M) \geq 1 - \frac{5}{16L}, \quad \cos^L(\pi/M) \geq 1 - \frac{5}{16} = \frac{11}{16}, \quad \rho_N \leq \frac{16\sqrt{2}}{110} < 1.$$

The first bound uses $M \geq 4\sqrt{L}$ and $\pi^2 < 10$; the second is Bernoulli's inequality.

Initialize by a convex decomposition of the observed seed ax in P_0 . At cut t , the machine stores $s \in [M]$. After command c , replace it by a random label whose mean vertex in P_{t+1} is $R^c \rho_t u_s$, using (10). Induction gives

$$\mathbb{E}[\rho_t u_{s_t} \mid x, c_1, \dots, c_t] = R^{c_1 + \dots + c_t} ax.$$

The terminal binary row has mean $e_j^T \rho_N u_s$. This proves the specified full conditional law. The final binary register fits within M states.

The update coefficients can be written explicitly. If the direction of $R^c u_s$ falls between u_l and u_{l+1} , let its offset be $0 \leq \theta \leq \omega = 2\pi/M$. Set

$$A = \frac{\rho_t \sin(\omega - \theta)}{\rho_{t+1} \sin \omega}, \quad B = \frac{\rho_t \sin \theta}{\rho_{t+1} \sin \omega}, \quad T_{t,c}(s, k) = A \mathbf{1}_{k=l} + B \mathbf{1}_{k=l+1} + \frac{1 - A - B}{M}.$$

Both coefficients are nonnegative and $A + B \leq (\rho_t/\rho_{t+1}) \sec(\pi/M) = 1$. Since $\sum_k u_k = 0$, these rows have the required mean. Boundary choices agree. Initialization uses the same triangular fan, replacing ρ_t by the norm of the seed. Only a vertex *index* is stored. In fact, the ratio ρ_t/ρ_{t+1}

is constant, so each command matrix is the same at every training epoch for this construction. The final decoder still depends on the chosen horizon.

For the rational alternative, decompose ax into the three vertices v_i of (8); its weights are

$$1/2 + ax_1, \quad 1/4 - ax_1/2 + ax_2, \quad 1/4 - ax_1/2 - ax_2.$$

Store the pair (i, q) , initially $q = 0$, and add c to q at each command. At cut t at most $3(t + 1)$ states are available. The decoder has mean $e_j^T R^q v_i$, which belongs to $[-1, 1]$ because $\|v_i\|_2 < 1$. This is an exact realization, with rational initialization, transitions and output rows when R is rational. $\square$

The polygon construction provides known exact stochastic rows. For a rational rotation, its trigonometric constants are algebraic; for the golden angle they are fixed computable real constants. Neither is being silently identified with a bounded fair-bit computation. The rational $3(N + 1)$ construction can use the separately charged rational-row sampler of the retained development [13]. Its extra workspace is not asserted to preserve the atomic peak. Similarly, an autonomous machine required to check the input length has an additional clock task; that task is not part of Definition 2.1.

5. AN EXPLICIT MULTI-CUT INEQUALITY

We now give a quantitative expansion estimate for the rotation in Theorem 1.1. Throughout this section, $\alpha = (\sqrt{5} - 1)/2$, R is rotation by $2\pi\alpha$, and $a = 1/10$. The Euclidean unit disk is denoted by $\mathbb{B}^2$.

Lemma 5.1 (Finite orbit covering). *For every integer $M \geq 3$, there is an integer q with $32M \leq q < 64M$ such that $\{R^k u : 0 \leq k < q\}$ is a $3\pi/q$ -net of the unit circle in angular distance, for every unit u .*

Proof. Let $F_0 = 0, F_1 = 1, F_{m+1} = F_m + F_{m-1}$ and take the first Fibonacci number $q = F_m$ at least $32M$. Consecutive positive Fibonacci numbers have ratio less than two, so $q < 64M$. Put $p = F_{m-1}$. The elementary identities

$$\gcd(p, q) = 1, \quad F_m \alpha - F_{m-1} = (-1)^{m-1} \alpha^m, \quad F_m \alpha^m < 1$$

follow by induction (or by the formula for F_m obtained from the two roots of $z^2 = z + 1$). Thus $|\alpha - p/q| < 1/q^2$. The first q multiples of p/q form the uniform q -grid modulo one. Each differs from the corresponding multiple of α by less than $1/q$ in circular distance. Every point of the circle is within $1/(2q)$ of a grid point and hence within $3/(2q)$ turns of an orbit point. Multiplying by 2π proves the assertion, also after a common rotation by u . $\square$

Lemma 5.2 (A polygon cannot be too nearly invariant). *If $a\mathbb{B}^2 \subset P \subset \sqrt{2}\mathbb{B}^2$ is a polygon with at most $M \geq 3$ vertices, then*

$$(11) \quad d_H(P, RP) \geq \frac{a}{64M^3}.$$

Proof. Write $\varepsilon = d_H(P, RP)$ and $r = \max_{z \in P} \|z\|_2$. Choose a vertex v of norm r . Rotational invariance of Hausdorff distance and the triangle inequality give $d_H(P, R^k P) \leq k\varepsilon$. By Lemma 5.1, for every unit direction u some $k < q$ puts $R^k v$ within angular distance $3\pi/q$ of u . For the support function h_P , it follows that

$$(12) \quad h_P(u) \geq r \cos(3\pi/q) - q\varepsilon, \quad r - h_P(u) \leq \frac{9\pi^2 \sqrt{2}}{2q^2} + q\varepsilon.$$

The directions of the at most M vertices have an empty circular arc of length at least $2\pi/M$. At its midpoint u_0 , every vertex has scalar product at most $r \cos(\pi/M)$. Consequently

$$r - h_P(u_0) \geq r(1 - \cos(\pi/M)) \geq \frac{2a}{M^2}.$$

Here $1 - \cos s \geq 2s^2/\pi^2$ for $0 \leq s \leq \pi$, and $r \geq a$. Using $q \geq 32M$, $\pi^2 < 10$ and $\sqrt{2} < 3/2$ shows that the first term on the right of (12) is less than a/M^2 . It follows that $q\varepsilon \geq a/M^2$. The bound $q < 64M$ proves (11). $\square$

Lemma 5.3 (Area pays for Hausdorff displacement). *If $a\mathbb{B}^2 \subset P \subset R_*\mathbb{B}^2$ and $U = \text{conv}(P \cup RP)$, for an arbitrary planar rotation R , then*

$$(13) \quad \text{area}(U) - \text{area}(P) \geq \frac{a}{R_*} d_H(P, RP)^2.$$

Proof. Let $d_+ = \max_{u \in \mathbb{S}^1} (h_{RP}(u) - h_P(u))_+$. When $d_+ > 0$, choose a maximizing direction u and $x \in RP$ with $u \cdot x = h_P(u) + d_+$. The triangle with vertices x and the two points $\pm av$, where $v \perp u$ is a unit vector, lies in U . Its part in $\{z : u \cdot z > h_P(u)\}$ lies outside P . Similar triangles give its area as

$$\frac{ad_+^2}{u \cdot x} \geq \frac{ad_+^2}{R_*}.$$

The zero case needs no argument. Repeating this with P and RP interchanged gives the same bound with d_- . Rotation preserves area, so the two excess areas coincide. For compact convex sets, $d_H(P, RP) = \max(d_+, d_-)$, proving the claim. $\square$

Theorem 5.4 (Quantitative temporal obstruction). *For the golden-angle rotation and S, D of Section 4,*

$$(14) \quad \delta_M \geq \frac{1}{6000000M^6} \quad (M \geq 3).$$

Every exact realization of $C_{N,R}$ therefore satisfies

$$(15) \quad \sum_{t=0}^{N-1} 2^{-6K_t} \leq 24000000.$$

In particular $W_N(R) \geq \frac{1}{6} \log_2(N/24000000)$.

Proof. A polygon feasible for (6) contains $a\mathbb{B}^2$ and lies in $\sqrt{2}\mathbb{B}^2$. The last two lemmas give

$$\text{area}(\text{conv}(P \cup RP)) - \text{area}(P) \geq \frac{a^3}{4096\sqrt{2}M^6} > \frac{1}{6000000M^6},$$

using $a = 1/10$ and $\sqrt{2} < 10/7$. Minimize over P to obtain (14). Apply Theorem 3.2 and bound $\text{area}(D) - \text{area}(S)$ by 4. If $K_t \leq W$ at all times, the left side of (15) is at least $N2^{-6W}$. $\square$

Combining Proposition 4.1, Theorems 4.2, 4.3 and 5.4 proves Theorem 1.1. The profile inequality gives more than a bound on one minimum peak. For example, at most $24000000 \cdot 2^{6k}$ of the command-input cuts can have width at most k , independently of the widths at the other cuts. The same statement follows directly by restricting the nonnegative sum.

Proposition 5.5 (Robustness at a fixed excluded profile). *Fix a finite horizon and a profile excluded for $C_{N,R}$ by either volume bound. There is a strictly positive uniform row-total-variation distance from $C_{N,R}$ to all arrays realized with that profile. For the rational family, a positive rational lower distance can be computed from the fully listed array and profile by real quantifier elimination.*

Proof. Initialization, update and decoder rows range over a finite product of compact simplices. The realized array is a polynomial, and hence continuous, function of those rows. Its image is compact. An excluded array has strictly positive distance to this image in the maximum of the finitely many row-TV distances. For rational target entries the minimum distance is a semialgebraic optimization, after expressing absolute values by auxiliary variables; real quantifier elimination and root isolation give the stated rational lower bound [2]. No uniform lower distance as the horizon or profile varies follows from this argument. $\square$

The exact golden-angle lower bound uses a specified computable real rotation; it is not presented as an algebraic-input computation. The rational family supplies algebraic excluded-horizon and fixed-profile robustness certificates instead. Neither the numerical values of its δ_M nor optimal rates are asserted by the finite regression tests.

6. EXACT COMPOSITION OF FEASIBLE PROFILES

The previous bounds accumulate a compatibility cost through time. There is also an exact composition operation across distinguished experiments. Its hypothesis is operational: the distinguishing label must be recovered at the end and is not externally available after its initial preparation.

Take m complete terminal arrays C_i , on a common horizon and common future input alphabets. Their initial alphabets may be identified by padding. Define the *verifiably tagged sum* $C^\oplus$ by receiving (i, x) initially and requiring terminal output (i, b) with law $C_i(b \mid x, \mathbf{y}, j)$. A wrong tag has probability zero. The tag is not repeated in later inputs, and previous outputs are not readable as memory. All branches have the same full suffix alphabet.

For two sets of integer profiles their sum below denotes coordinatewise Minkowski sum, including the already defined upward closure.

Theorem 6.1 (Profile addition under a retained tag). *For the internal registers of the tagged sum,*

$$(16) \quad \mathcal{F}(C^\oplus) = \mathcal{F}(C_1) + \cdots + \mathcal{F}(C_m).$$

The identity holds for arbitrary horizons and arbitrary stochastic realizations. It is an identity of complete feasible-profile sets, not just of separate minima or a scalar matrix rank.

Proof. Fix a realization and a cut t . Let $S_{t,i}$ consist of states having positive probability after some prefix from branch i . If one state belonged to $S_{t,i}$ and $S_{t,k}$ with $i \neq k$, run the same fixed future input word from that state. Exact tag recovery in branch i forces this conditional continuation to emit tag i with probability one: a positive mixture weight could not otherwise give zero wrong-tag probability. The same reasoning in branch k forces tag k . This is impossible. Thus the sets $S_{t,i}$ are disjoint.

For every allowed next input, a positive transition from $S_{t,i}$ leads to $S_{t+1,i}$, since its source is reachable after some branch- i prefix and the full product suffix is allowed. Restricting the machine to these state sets and stripping its terminal tag therefore gives a complete realization of C_i . In particular its profile $(|S_{t,i}|)_t$ is feasible. Their coordinatewise sum does not exceed the original profile. Padding any one of the restricted realizations absorbs each unused coordinate and gives membership in the right side of (16).

Conversely, use disjoint copies of the branch registers and transition rows. Initialization selects the declared branch, which is thereafter stored by membership in its disjoint register. The final decoder emits its tag and answer. This gives the sum of any feasible branch profiles and proves the reverse inclusion. No independent shared seed or free branch selector is supplied. $\square$

If the final tag were dropped, this proof would fail: states serving two branches could be identified. Likewise, branch-dependent legal suffixes must not be substituted for the common full alphabet without a separate argument. These are hypotheses of the operation, not conclusions about unlabelled mixtures.

Corollary 6.2 (An exact growing two-cut frontier). *Let every branch be the two-cut array (18), with a fixed $9/10 \leq h < 1$, and let $m \geq 1$. Its tagged sum has exactly the feasible internal profiles*

$$(17) \quad K_1 \geq 3m, \quad K_2 \geq 3m, \quad K_1 + K_2 \geq 7m.$$

The Pareto-minimal pairs are $(3m + j, 4m - j)$, $j = 0, \dots, m$. Both separate minima are $3m$, and the minimum full peak, including the $2m$ -symbol terminal output, is $\lceil 7m/2 \rceil$.

Proof. Appendix A proves the primitive feasible set as the upward closure of $(3, 4), (4, 3)$. Its m -fold Minkowski sum has the listed minimal pairs. To check that (17) has no additional holes, put $u = K_1 - 3m$, $v = K_2 - 3m$. If $u, v \geq 0$ and $u + v \geq m$, choose an integer j with $\max(0, m - v) \leq j \leq \min(m, u)$. Then $(K_1, K_2) \geq (3m + j, 4m - j)$. The peak is minimized by the nearest integer balance of j and $m - j$, giving $\lceil 7m/2 \rceil$. The output register has $2m \leq \lceil 7m/2 \rceil$ labels. $\square$

The primitive answer probabilities have full support, but the tagged sum has zeros at wrong tags. We do not use it as the full-support part of Theorem 1.1; that theorem already has a uniform full-support interval and fixed alphabets. The two results answer different questions: (17) is an exact finite Pareto law, whereas (15) excludes bounded simultaneous width while every separate positive minimum remains three.

7. COMPARISON, IMPLEMENTATION AND FURTHER QUESTIONS

7.1. Stationary positive realization and finite causal width. Benvenuti–Farina [1, Theorems 2 and 3] formulate existence of positive realizations by invariant polyhedral cones and shift-invariant positive sequences. Their Example 4 uses an irrational planar rotation to prevent a finite stationary realization. Monràs–Winter [8, Theorem 6 and Lemma 7] give the corresponding stochastic cone criterion and a rotation example. These are direct antecedents, not merely neighboring subjects.

Theorem 1.1 makes a different finite assertion. The matrices are allowed to be unrelated at different epochs, and a new machine may be selected at every horizon. Each entire finite array is positively realizable, while each of its cuts separately needs exactly three states. The lower bound is the accumulation of enlargement along $P_t \subset P_{t+1}$ and $RP_t \subset P_{t+1}$. An invariant-cone existence theorem or a peripheral-spectrum argument for one common matrix does not, by itself, give this quantitative inequality for changing rows. Conversely, our theorem does not re-prove or extend the full stationary positive-realization existence classification.

The reachable-section construction is deliberately compatible with classical quotient realizations. In [8, Section 4, Theorem 9], the accessible subspace and its unobservable kernel determine the minimal linear quotient; arbitrary positive realizations can contain more directions. That is why Lemma 3.1 uses a section of a simplex before projecting. It does not identify minimum hidden dimension with a minimum number of vertices of an unlifted observable polygon. The related distinction between restricted and unrestricted positive factorizations appears explicitly in [5, Definition 1, Theorem 1]. The exponential allowance 2^{K_t} is conservative, but makes the lower bound valid for all competing hidden representations.

The recent work of Lumbreras–Ma–Thompson–Gu [7] studies irreducible errors and decision failures of finite classical world models, with exact quantum realizations. Its Section IV.2 and ensuing proofs use repetition of one update, Perron–Frobenius limits and an irrational phase.

Our task has no quantum simulator or decision-loss claim: it compares exact *finite-horizon* cut-dependent classical realizations. The common rotation ancestry and different row-sharing quantifiers are both relevant. We do not infer a novelty boundary from the absence of a citation to this manuscript in that work.

7.2. Residual restrictions, direct sums and sampling. The finite-generation and canonical residual constructions of Esposito–Lemay–Denis–Dupont [4] and Denis–Esposito [3, Proposition 16] concern state languages constrained by the residual representation. The lower bound here places no such constraint on a state’s full continuation. Indeed all separate minima stay three in the rotation example, so a large list of actual residuals alone would not prove its causal width lower bound. Classical hidden-Markov and filtered-experiment comparison remain part of the surrounding theory [12, 9]; our wordwise finite profile statement is not claimed to be a new general comparison theorem for filtered spaces.

Nonnegative direct-sum additivity itself is familiar, including the tensor version in [11, Lemma 4.1]. Theorem 6.1 uses the same positivity principle to separate *reachable state sets at every time*; it identifies a sum of entire feasible profile sets. Taking only sums of separate ranks would lose the tradeoff in (17). Its tag and full common suffix alphabet are explicit and cannot be omitted.

The additive encoder construction in the preserved article [13] uses weighted replacement. The sampling step is weighted reservoir sampling, as described, for example, in [10, Appendix A.2]: replace the current sample with probability equal to the new weight divided by total weight. When total previous weight is zero, the dummy label is chosen from the existing alphabet. Neither that sampling algorithm nor the classical simplex inputs are novelty claims of the present paper. The earlier online consequences and rational sampler remain available without being needed for the rotation lower bound.

7.3. What is and is not charged. The finite-horizon bounds concern exact atomic stochastic transducers, not a bit-operation model. Both alphabet cardinality and all persistent hidden choices are charged. The row table and the prescribed external epoch are not. The rational family permits separately charged fair-bit sampling; the algebraic or computable-real polygon rows are not given a finite-denominator implementation claim. Sampling and verification costs therefore do not disappear into the phrase “exact realization.”

The new lower bound holds even with the external clock and arbitrary cut-dependent rows. The polygon upper construction has common command matrices across its training epochs, but its prescribed decoder depends on N . This does not define one finite machine that works at every horizon with an unchanged decoder. No automatic transfer from the clocked peak to autonomous length recognition is made.

7.4. Conclusions. The proved separation is between *separate positive minima* and *simultaneous positive width*, with fixed alphabets, uniformly positive response probabilities, and an arbitrarily long horizon. The projected reachable-section budget applies to a class of volume-preserving mean dynamics. The golden-angle family has an explicit multi-cut inequality and a constructive square-root upper bound; the rational family has effective excluded-horizon certificates. The tagged operation separately supplies exact growing Pareto frontiers.

The optimal growth of W_N , a sharp measure of lift complexity, quantitative stability uniform in the horizon, and general optimization of informative acquisition or validation actions remain distinct questions. The logarithmic lower bound is not matched by the square-root upper bound. Complete arbitrary-profile minimization is not claimed. These distinctions do not affect the unbounded separation or the exact tagged frontier.

The historical geometric, large-deviation and nonlinear-operator chains are preserved as separate mathematics. No finite-array argument here is claimed to discharge their analytic gates. The focused article contains the proofs required for its new claims; the cumulative volumes are archival records, not an additional burden of certification for its referee.

APPENDIX A. THE RETAINED TWO-CUT PRIMITIVE

The following proof is retained from revision 31 [13]; its role here is the input to Corollary 6.2, not a new claim.

The following experiment has only two internal memory cuts. The first input is $x = (x_1, x_2) \in \{-1, 1\}^2$, received as one symbol. A command $j \in \{1, 2, 3, 4, 5\}$ is then received, followed by a query $k \in \{1, 2\}$. The machine emits $B \in \{-1, 1\}$ in the update receiving k . Let K_1 count states after x and K_2 states after j ; a retained final answer has two labels. No further access to x or j is permitted. For $0 < h \leq 1$ prescribe the conditional mean vectors

$$(18) \quad \begin{array}{c|ccccc} j & 1 & 2 & 3 & 4 & 5 \\ \hline z_{x,j} & (h, h) & (h, -h) & (-h, 0) & (-h/2, hx_1/4) & (-h/2, hx_2/4) \end{array}.$$

The answer law is $\mathbb{P}(B = b \mid x, j, k) = (1 + bz_{x,j}(k))/2$.

The program is clocked and uses arbitrary stochastic rows. Thus every allowed implementation has common rows $E(s \mid x)$, $T(t \mid s, j)$ and $D(b \mid t, k)$; all past randomness is in s or t .

We first record the geometric estimate used by the converse.

Lemma A.1 (A quantitative forced triangle). *Put $h = 1 - \delta$, where $0 \leq \delta \leq 1/10$. If a triangle $P \subset [-1, 1]^2$ contains (h, h) , $(h, -h)$ and $(-h, 0)$, then every point $(-h/2, y) \in P$ satisfies*

$$(19) \quad |y| \leq \beta(\delta) := \frac{1}{4} + \frac{\delta}{2(1-\delta)} + \frac{\delta}{2-3\delta} < \frac{h}{2}.$$

Proof. A convex representation of (h, h) includes a vertex A for which $2 - A_1 - A_2 \leq 2\delta$, so $A_1, A_2 \geq 1 - 2\delta$. A representation of $(h, -h)$ similarly includes a vertex B with $B_1 \geq 1 - 2\delta$ and $B_2 \leq -1 + 2\delta$. These vertices are distinct. Call the third vertex D . In a representation of $(-h, 0)$, let u be the total weight on A, B . Since their first coordinates are at least $1 - 2\delta$ and $D_1 \geq -1$,

$$-1 + \delta \geq -1 + 2(1 - \delta)u, \quad u \leq \frac{\delta}{2(1 - \delta)}.$$

The zero second coordinate implies $|D_2| \leq u/(1 - u) \leq \delta/(2 - 3\delta)$. For a point of P with first coordinate $-h/2$, the total weight v on A, B satisfies

$$-h/2 \geq -1 + 2hv, \quad v \leq \frac{1 - h/2}{2h} = \frac{1}{4} + \frac{\delta}{2h}.$$

Its second coordinate has absolute value at most $v + (1 - v)|D_2| \leq \beta(\delta)$. Finally the two fractions in β increase with δ while $h/2$ decreases, and

$$\beta(1/10) = \frac{1}{4} + \frac{1}{18} + \frac{1}{17} = \frac{223}{612} < \frac{9}{20}.$$

This proves the strict bound throughout the interval, including zero. $\square$

Lemma A.2 (Centered square in an inscribed triangle). *If a triangle in $[-b, b]^2$ contains $[-a, a]^2$, with $a > 0$, then $b \geq 2a$.*

Proof. Write its three barycentric functions as $\lambda_s(u) = c_s + d_s \cdot u$. They are nonnegative on $[-a, a]^2$, so $c_s \geq a\|d_s\|_1$ and $\sum_s c_s = 1$. If the vertices are v_s , affine reconstruction gives $\sum_s v_s d_s^T = I_2$. Taking traces yields $2 \leq b \sum_s \|d_s\|_1 \leq b/a$. $\square$

This necessary bound is sufficient for our contradiction; no claim of its attainability in dimension two is used. The required two-dimensional trace argument has been proved here; its general-dimensional geometric background is recorded in [13].

Theorem A.3 (Exact incompatible minima and their Pareto frontier). *For every $9/10 \leq h \leq 1$, both cuts of (18) separately admit an exact three-state realization and have ordinary normalized residual rank three. No realization has $(K_1, K_2) = (3, 3)$. The entire feasible profile region is*

$$(20) \quad \{(K_1, K_2) : K_1 \geq 3, K_2 \geq 4\} \cup \{(K_1, K_2) : K_1 \geq 4, K_2 \geq 3\}.$$

Thus the exact Pareto-minimal profiles are $(3, 4)$ and $(4, 3)$, and the minimum full peak is four. For $h < 1$ the specified array has full support. At $h = 9/10$ all answer probabilities belong to $[1/20, 19/20]$.

Proof. The four rows at the first cut vary independently through the last two second coordinates of (18). They form a nondegenerate affine square, hence have normalized row rank three. At the second cut the response vectors lie in $\mathbb{R}^2$ and include three affinely independent points $(h, h), (h, -h), (-h, 0)$. Its normalized row rank is also three. Factoring through a stochastic register cannot increase rank, so both state counts are at least three.

Here are matching one-sided constructions. Put $a = h/4$. At the first cut, three continuation vectors with free coordinate pairs

$$(21) \quad (-a, -a), \quad (3a, -a), \quad (-a, 3a)$$

and the fixed coordinates from (18) enclose the four actual rows. The triangle is legal because $3a \leq 3/4$. The barycentric encoder weights for (ax_1, ax_2) are

$$(\lambda_0, \lambda_1, \lambda_2) = \left(\frac{2 - x_1 - x_2}{4}, \frac{1 + x_1}{4}, \frac{1 + x_2}{4} \right).$$

After j , represent the resulting two-dimensional mean in the convex hull of the four corners of $[-1, 1]^2$. For a mean (u, v) the weights on $(\varepsilon_1, \varepsilon_2)$ are $(1 + \varepsilon_1 u)(1 + \varepsilon_2 v)/4$. The decoder returns the queried corner coordinate. This is a single machine of profile $(3, 4)$.

For profile $(4, 3)$, first retain x . Every actual second-cut mean lies in the triangle with vertices $h(1, 1), h(1, -1), h(-1, 0)$. At a mean (u, v) its weights on those vertices are

$$(22) \quad \frac{h + u + 2v}{4h}, \quad \frac{h + u - 2v}{4h}, \quad \frac{h - u}{2h}.$$

For the last two columns they are respectively $(1/4, 0, 3/4)$ or $(0, 1/4, 3/4)$; for the first three columns they are the three unit vectors. These rows followed by the binary decoder prove the second construction. All tables are rational when h is rational.

Suppose now that a $(3, 3)$ machine exists. Let G_1, G_2, G_3 be its full first-cut continuation arrays. Since their convex hull contains an affine square of dimension two, their affine span equals the affine span of the four actual rows. Therefore each G_s has the first three columns of (18) unchanged and has last columns $(-h/2, u_s)$ and $(-h/2, v_s)$ for some u_s, v_s . The first encoder then forces

$$(23) \quad [-a, a]^2 \subseteq \text{conv}\{(u_s, v_s) : 1 \leq s \leq 3\}.$$

This affine-slice deduction allows arbitrary state continuations; it neither assumes that they are residuals nor restricts them beforehand to the residual hull.

The three states at the second cut have decoder mean vectors $D_1, D_2, D_3 \in [-1, 1]^2$. Their triangle P contains all actual second-cut rows, and hence the three points in Lemma A.1. Every shifted continuation of each G_s must also lie in P , because the update after j is a common stochastic row into these same three decoder states. In particular, $(-h/2, u_s), (-h/2, v_s) \in P$. Lemma A.1 gives $|u_s|, |v_s| \leq \beta(1 - h) < h/2 = 2a$. This contradicts Lemma A.2 applied to (23).

No $(3, 3)$ realization exists. Padding the two constructions proves (20). The full-support and probability bounds follow immediately from the displayed means. $\square$

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